MATRICES
(เมทริกซ์ ver. นอนตายดีกว่า)

$$\begin{array}{c} \xrightarrow{\text{Column}} \\ \text{Row}\downarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m\times n} \end{array}$$

Addition $\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} a+e & b+f \\ c+g & d+h \end{bmatrix}$ * ต้องเป็นmatrix ขนาดเดียวกัน only!

Multiplication SCALAR — $4\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 4a & 4b \\ 4c & 4d \end{bmatrix}$

MATRICES — $\begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}_{m\times n} \times \begin{bmatrix} g & h \\ i & j \end{bmatrix}_{n\times p} = \begin{bmatrix} ag+bi & ah+bj \\ cg+di & ch+dj \\ eg+fi & eh+fj \end{bmatrix}_{m\times p}$ * Column1 = Row2 only! (ถ้าไม่เท่ากัน = undefined)

PROPERTIES

- $A+B = B+A$
- $A+(B+C) = (A+B)+C$
- $(cd)A = c(dA)$
- $1A = A$
- $c(A+B) = c(A+B)$
- $(c+d)A = cA+dA$
- $(AB)C = A(BC)$
- $A(B+C) = AB+AC$
- $c(AB) = (cA)B = A(cB)$

IDENTITY MATRIX
เมทริกซ์เอกลักษณ์

$$\begin{bmatrix} 1 & 0 & \cdots & \cdots & 0 \\ 0 & 1 & \cdots & \cdots & 0 \\ 0 & \cdots & 1 & \cdots & 0 \\ 0 & \cdots & \cdots & \ddots & \vdots \\ 0 & \cdots & \cdots & 0 & 1 \end{bmatrix}$$

$AI_n = A$, $I_mA = A$ } เมื่อ A มีขนาด m×n

TRANSPOSE

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow A' = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$

$B = \begin{bmatrix} a \\ b \end{bmatrix}_{2\times 1} \rightarrow B' = \begin{bmatrix} a & b \end{bmatrix}_{1\times 2}$

PROPERTIES

- $(A^T)^T = A$
- $(A+B)^T = A^T + B^T$
- $(cA)^T = c(A^T)$
- $(AB)^T = B^TA^T$ *ระวังจำสลับที่กัน

INVERSE MATRIX A^{-1}

$AB = BA = I_n \rightarrow$ B เป็น inverse ของ A

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow A^{-1} = \frac{1}{\det A}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ ↳ for matrix 2×2 only!

for matrix n×n:

$A^{-1} = \frac{1}{\det A}(\operatorname{adj} A)$ $\quad \operatorname{adj}(A) = [C_{ij}]^T$, $C_{ij} = (-1)^{i+j}M_{ij}$

PROPERTIES

- $(A^{-1})^{-1} = A$ | $(A^T)^{-1} = (A^{-1})^T$
- $(A^k)^{-1} = A^{-1}\cdot A^{-1}\cdot \ldots \cdot A^{-1} = (A^{-1})^k$
- $(cA)^{-1} = \frac{1}{c}(A^{-1})$; $c \neq 0$ | $(AB)^{-1} = B^{-1}A^{-1}$

DETERMINANT *square matrix only!

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow \det A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$

$B = \begin{bmatrix} a & b & g \\ c & d & h \\ e & f & i \end{bmatrix} = \left[\begin{array}{ccc|cc} a & b & g & a & b \\ c & d & h & c & d \\ e & f & i & e & f \end{array}\right] = (adj + bhe + gcf) - (edg + fha + icb)$

$\underbrace{adj + bhe + gcf}$, $\overbrace{edg + fha + icb}$

$$\det A = |A| = \sum_{j=1}^{n} a_{1j}C_{1j} = a_{11}C_{11} + a_{12}C_{12} + \ldots + a_{1n}C_{1n}$$

PROPERTIES

- $\det(A) \neq 0$
- $\det(AB) = \det(A)\det(B)$
- $\det(cA) = c^n\det(A)$ n-ขนาดเมทริกซ์
- $\det(A^{-1}) = \frac{1}{\det(A)}$ | $\det(A) = \det(A^T)$

ถ้าเจอmatrix ที่มีลักษณะตามนี้ det = 0

$\begin{bmatrix} 0 & 0 & 0 \\ a & b & c \\ a & e & f \end{bmatrix}$ $\begin{bmatrix} a & b & 3a \\ d & e & 3d \\ g & h & 3g \end{bmatrix}$ $\begin{bmatrix} a & b & c \\ a & e & f \\ a & b & c \end{bmatrix}$

CRYPTOGRAPHY ถอดรหัสข้อความ

0 = __	7 = G	14 = N	21 = U
1 = A	8 = H	15 = O	22 = V
2 = B	9 = I	16 = P	23 = W
3 = C	10 = J	17 = Q	24 = X
4 = D	11 = K	18 = R	25 = Y
5 = E	12 = L	19 = S	26 = Z
6 = F	13 = M	20 = T	

M E E T _ M E _ M O N D A Y _

13 5 5 | 20 0 13 | 5 0 13 | 15 14 4 | 1 25 0

encode : [message] × [matrix]

decode : [message] × [inverse matrix]

CARMER'S RULE หาค่า x, y, z

ex. (2 ตัวแปร)

$\begin{matrix} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{matrix} \Rightarrow A = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix}, D = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$

$x = \dfrac{\det\begin{bmatrix} c_1 & b_1 \\ c_2 & b_2 \end{bmatrix}}{\det A}, \quad y = \dfrac{\det\begin{bmatrix} a_1 & c_1 \\ a_2 & c_2 \end{bmatrix}}{\det A}$

ex. (3 ตัวแปร)

$\begin{matrix} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \\ a_3x + b_3y + c_3z = d_3 \end{matrix} \Rightarrow A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, D = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$

$x = \dfrac{\begin{bmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{bmatrix}}{\det A}, \quad y = \dfrac{\begin{bmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{bmatrix}}{\det A}, \quad z = \dfrac{\begin{bmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{bmatrix}}{\det A}$

GAUSSAIN ELIMINATION

$\begin{matrix} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \\ a_3x + b_3y + c_3z = d_3 \end{matrix} \Rightarrow A = \begin{bmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{bmatrix} \xrightarrow{\text{ต้องทำออกมาเป็นmatrix ขั้นบันได}} \left[\begin{array}{ccc|c} 1 & \bullet & \bullet & \bullet \\ 0 & 1 & \bullet & \bullet \\ 0 & 0 & 1 & \bullet \end{array}\right] \rightarrow \begin{matrix} x + \bullet y + \bullet z = \bullet \\ y + \bullet z = \bullet \\ z = \bullet \end{matrix}$ } แก้สมการหา x, y, z

ขั้นบันได

GAUSSAIN-JORDAN

$\begin{matrix} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \\ a_3x + b_3y + c_3z = d_3 \end{matrix} \Rightarrow A = \begin{bmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{bmatrix} \xrightarrow{\text{ต้องทำออกมาเป็น identity matrix}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & \bullet \\ 0 & 1 & 0 & \bullet \\ 0 & 0 & 1 & \bullet \end{array}\right] \rightarrow \begin{matrix} x = \bullet \\ y = \bullet \\ z = \bullet \end{matrix}$